

Edeon — Registration Readiness Dossier

Compound: Mesotrione (CAS: 104206-82-8)

SMILES: CS(=O)(=O)c1ccc(C(=O)C2C(=O)CCCC2=O)c([N+](=O)[O-])c1

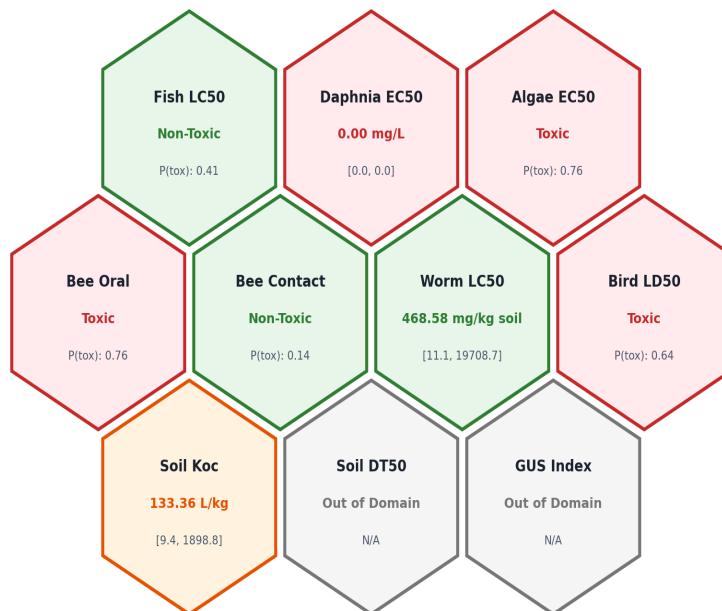
REGISTRATION READINESS VERDICT: NO_GO

Driver: Showstopper triggered: Groundwater concern (EU DWD 0.1 µg/L trigger), Aquatic hazard (CLP H400/H410/H411/H412), Soil persistence (EU 1107/2009) | **Confidence:** moderate

Rationale: GUS = 4.36 > 2.8 (leacher), GUS classification: leacher; Most sensitive endpoint: Daphnia EC50 = 0.0000 mg/L, Not rapidly degradable (default assumption); Predicted soil DT50 = 212 days, DT50 > 120 d — exceeds persistence cut-off criterion

Tier-1 Predictive Honeycomb

Mesotrione — QSAR Safety Profile



Regulatory Cut-Offs & Criteria Evaluation

Criterion / Endpoint	Status	Evidence & Details
Genotoxicity / Mutagenicity alerts (Benigni-Bossa)	WATCH	Aromatic nitro group (medium): Nitroreduction to reactive hydroxylamine/nitrene
Endocrine disruption screening (structural)	PASS	No endocrine disruption structural alerts detected
Skin sensitization alerts (protein reactivity)	PASS	No skin sensitization alerts detected
PBT/vPvB (REACH Annex XIII)	WATCH	Soil DT50 = 212 d > 180 d (vP); Daphnia EC50 = 0.0000 mg/L < 0.1 (T)
Groundwater concern (EU DWD 0.1 µg/L trigger)	LIKELY_SHO WSTOPPER	GUS = 4.36 > 2.8 (leacher); GUS classification: leacher
Aquatic hazard (CLP H400/H410/H411/H412)	LIKELY_SHO WSTOPPER	Most sensitive endpoint: Daphnia EC50 = 0.0000 mg/L; Not rapidly degradable (default assumption)
Acute mammalian toxicity (CLP oral)	PASS	Predicted rat oral LD50 = 5200 mg/kg
Soil persistence (EU 1107/2009)	LIKELY_SHO WSTOPPER	Predicted soil DT50 = 212 days; DT50 > 120 d — exceeds persistence cut-off criterion

IN-SILICO SCREENING ONLY — These results are computational triage signals based on predicted endpoints and structural pattern matching. They are NOT regulatory determinations and cannot replace experimental studies or expert regulatory assessment.